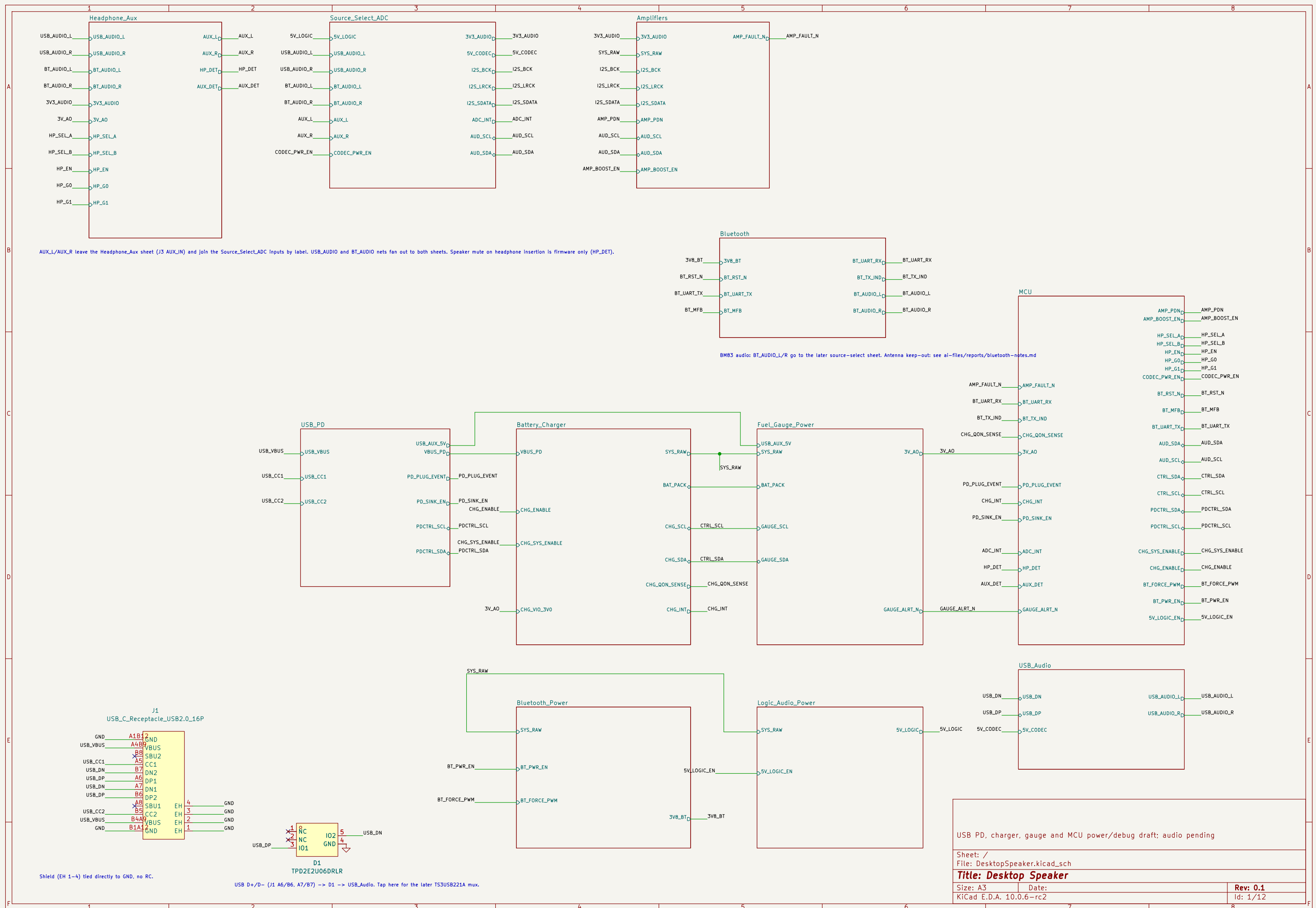
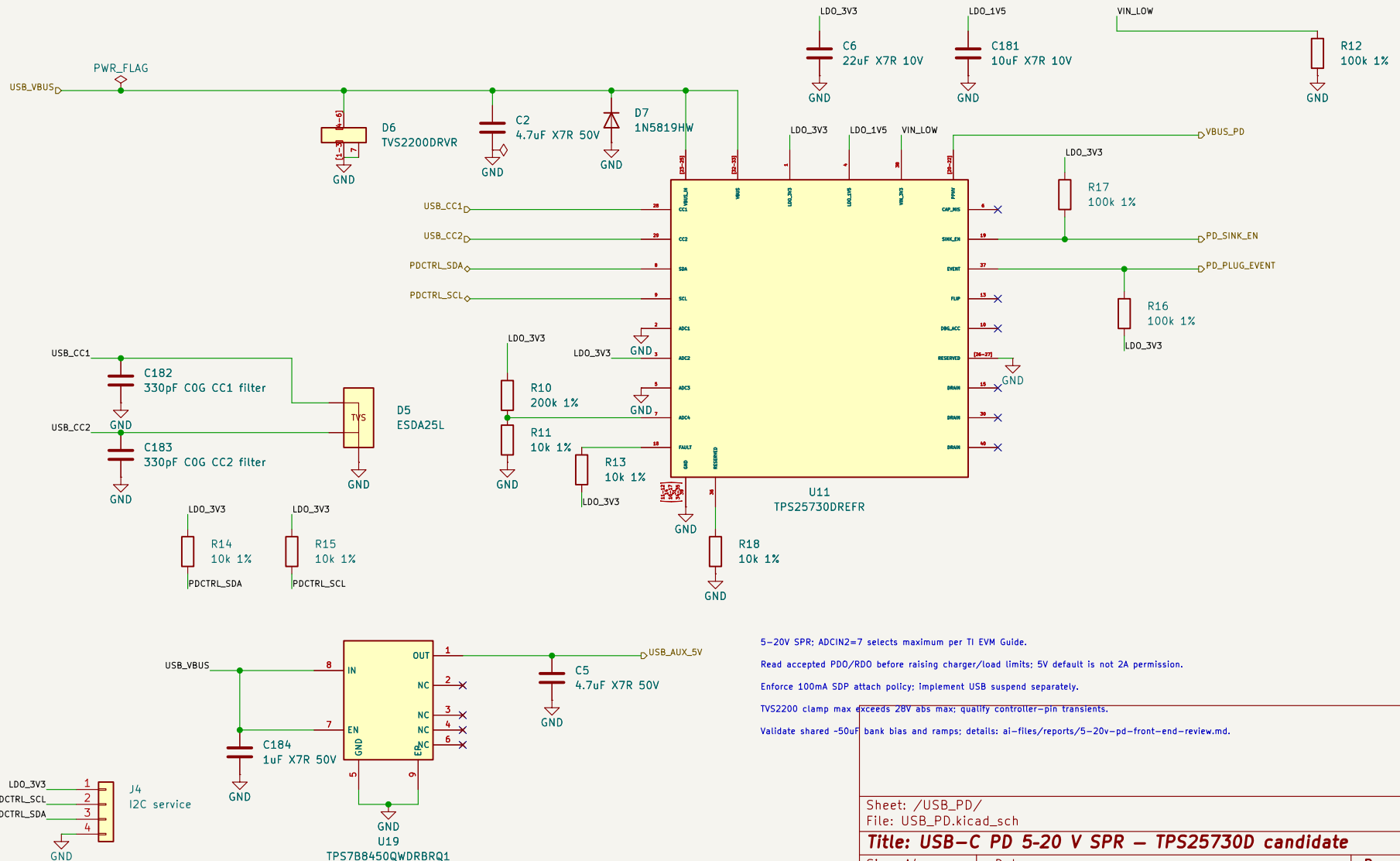






Sheet: /USB_PD/

File: USB_PD.kicad_sch

Title: USB-C PD 5-20 V SPR – TPS25730D candidate

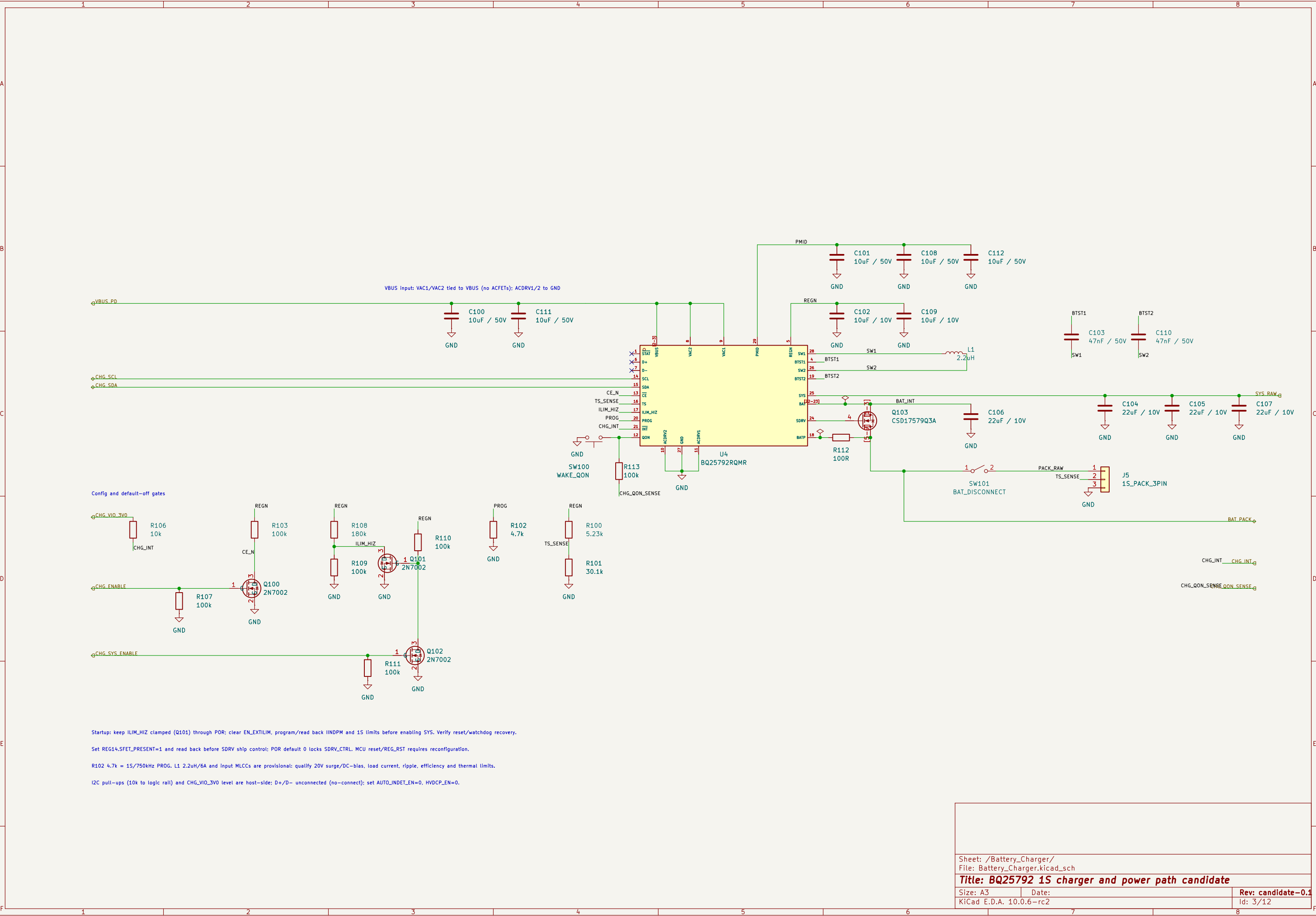
Size: A4

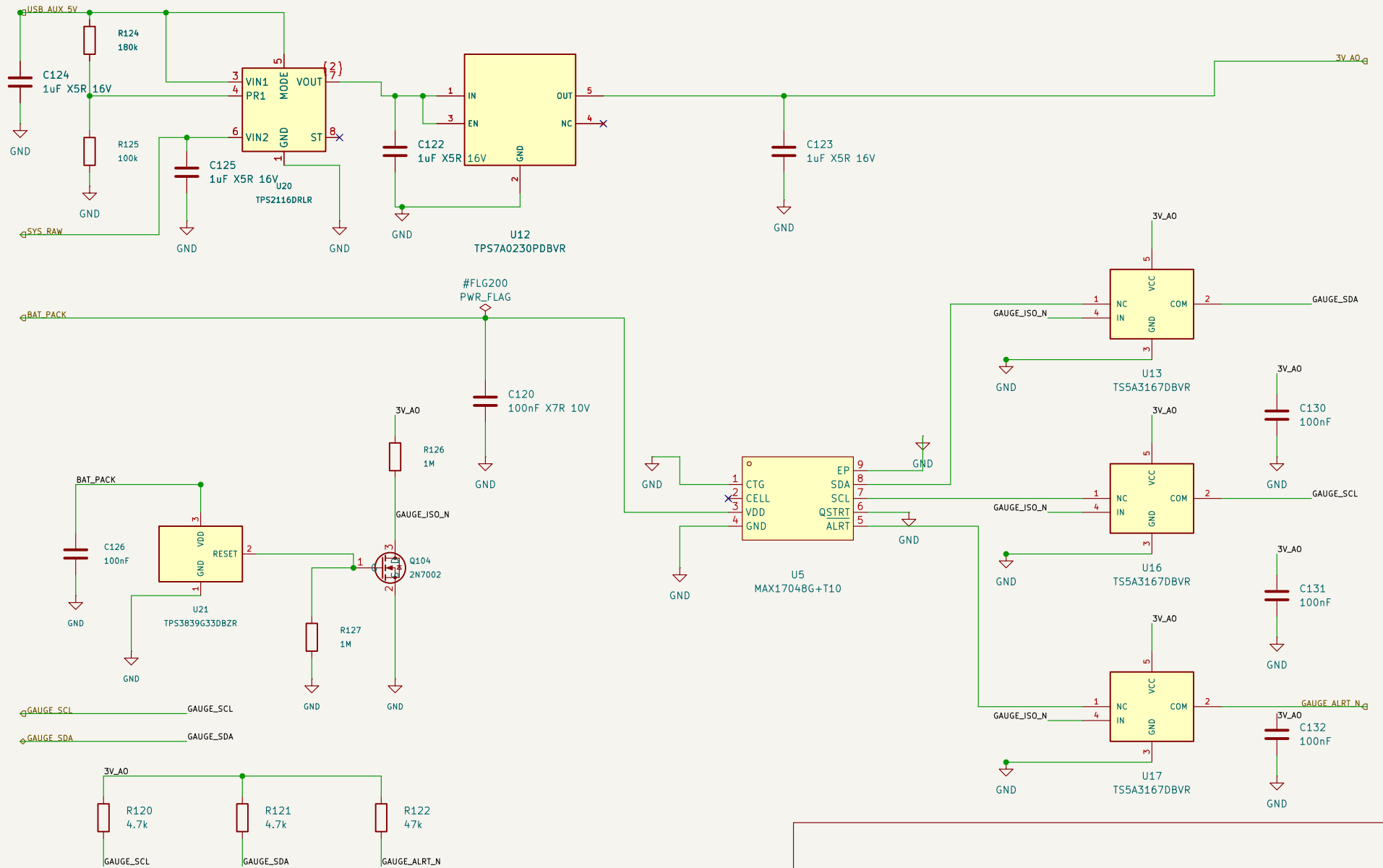
Date:

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Rev: prototype review

Id: 2/12





USB-priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus; AO-powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge_Power/
File: FuelGauge_Power.kicad_sch

Title: Fuel gauge and low-current power

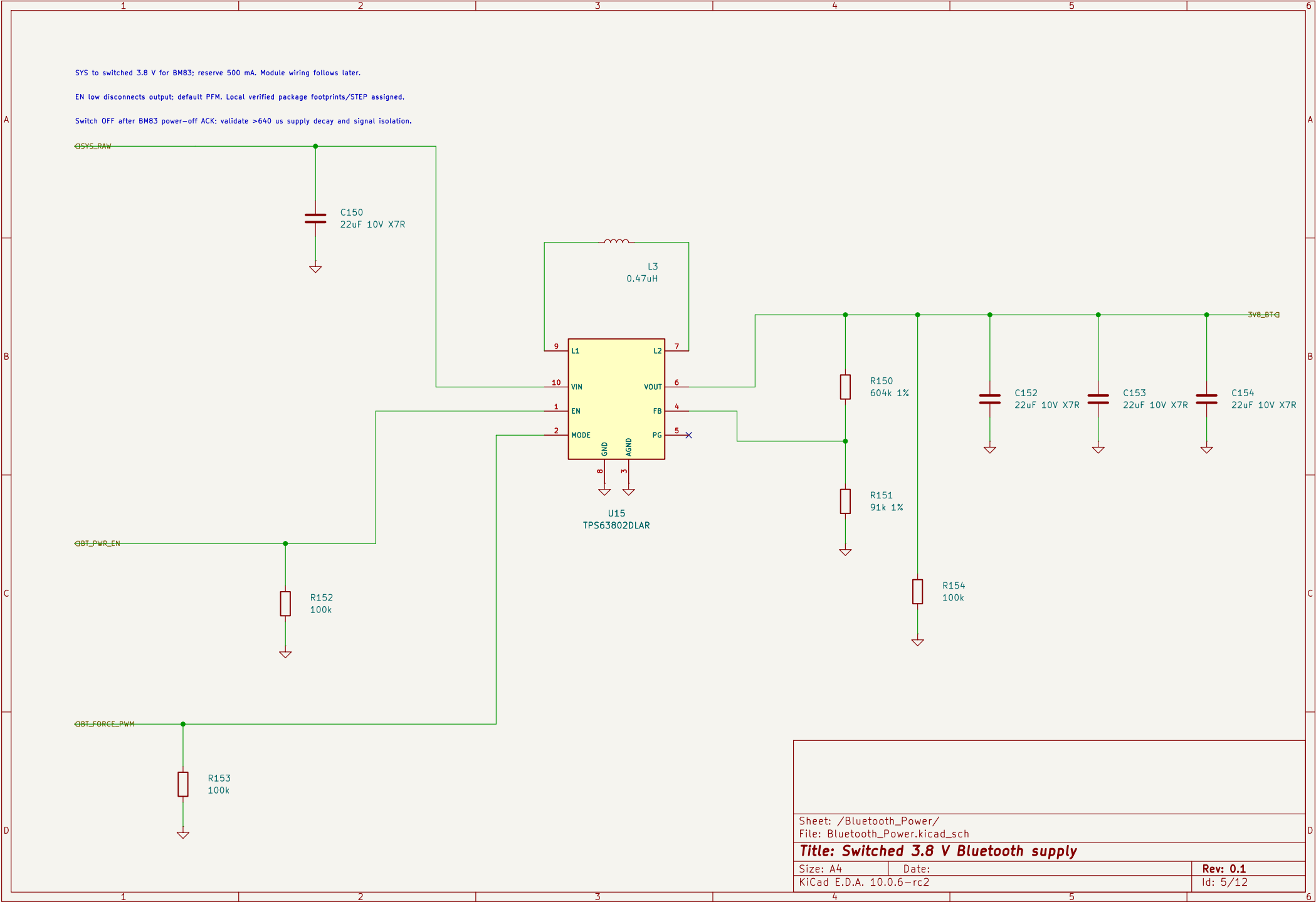
Size: A4

Date:

KiCad E.D.A. 10.0.6-rc2

Rev: 0.1

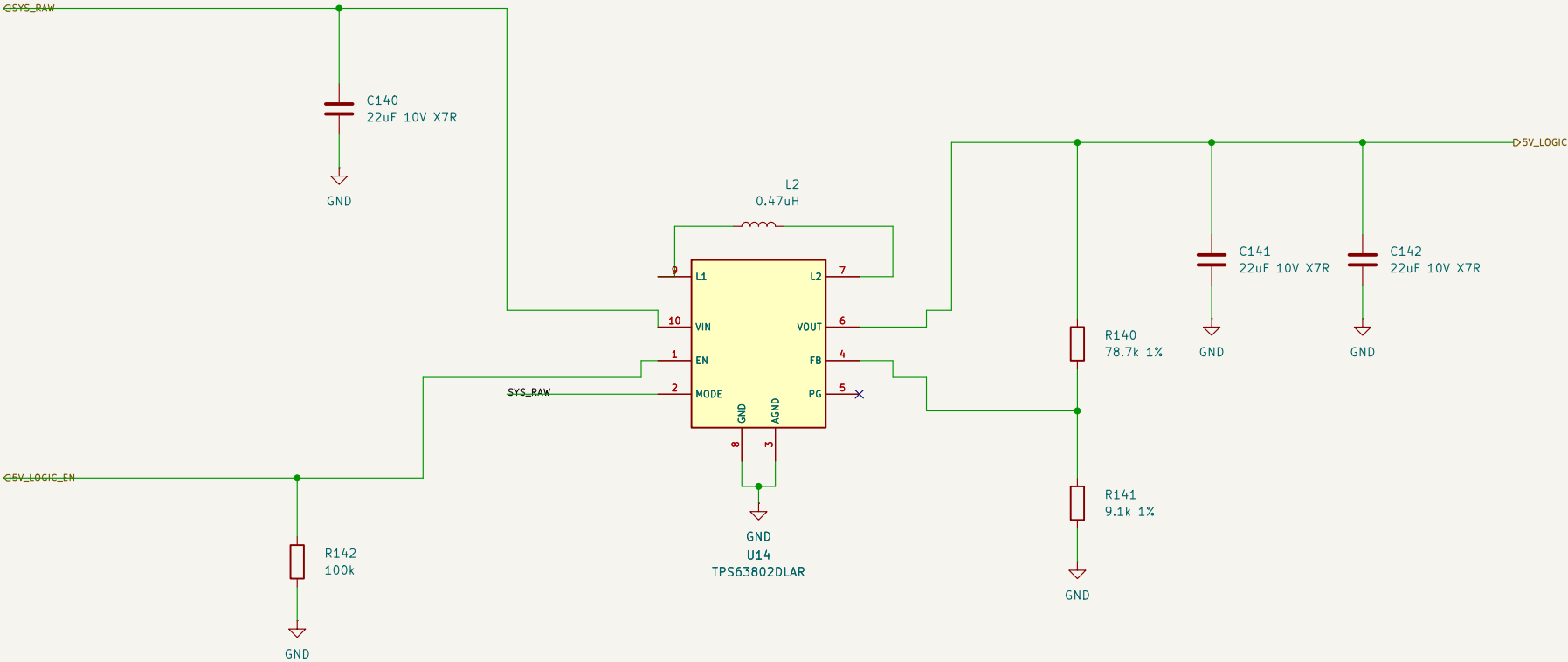
Id: 4/12



Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

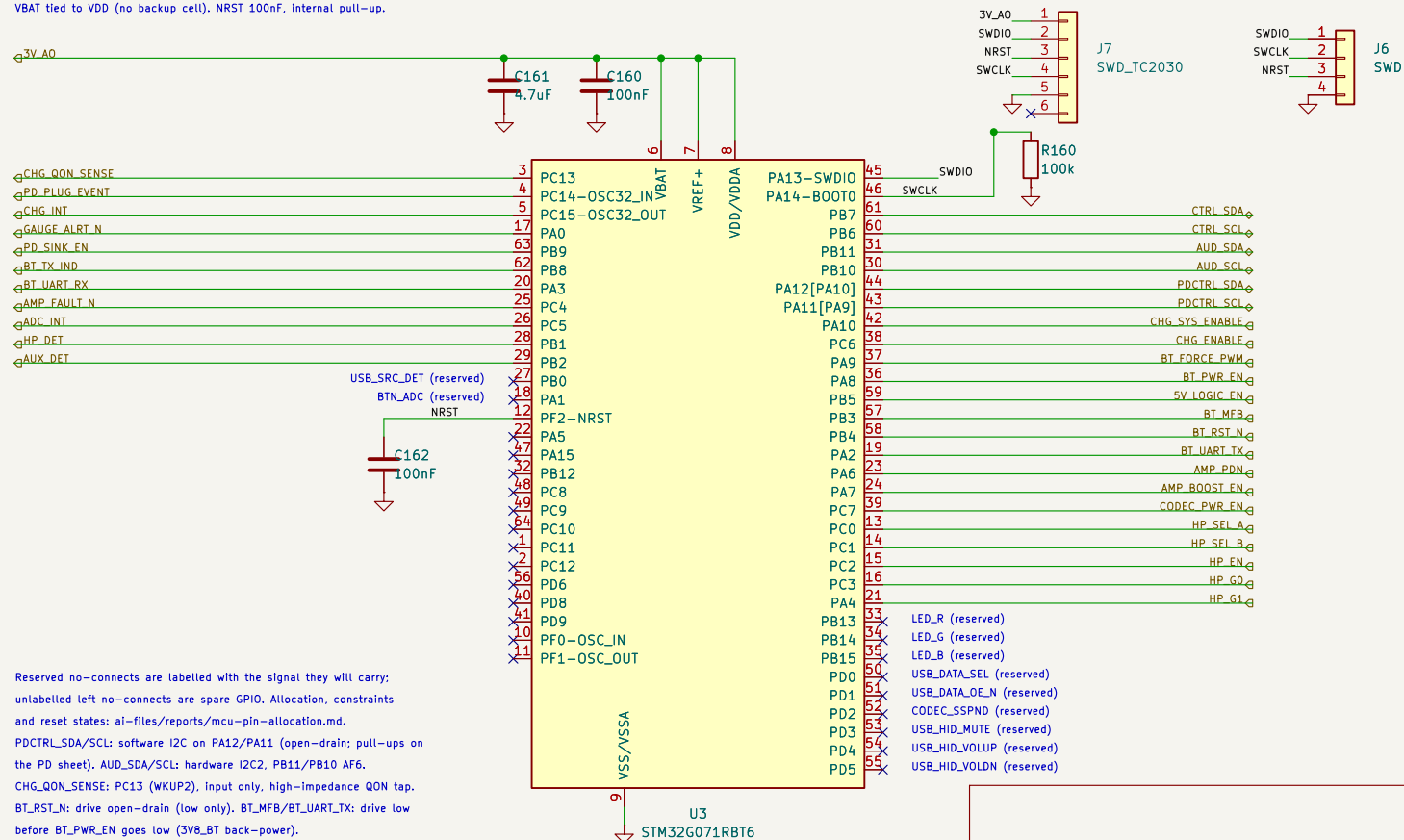
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ 100nF + 1uF (VREFBUF off, VREF+ = VDDA).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

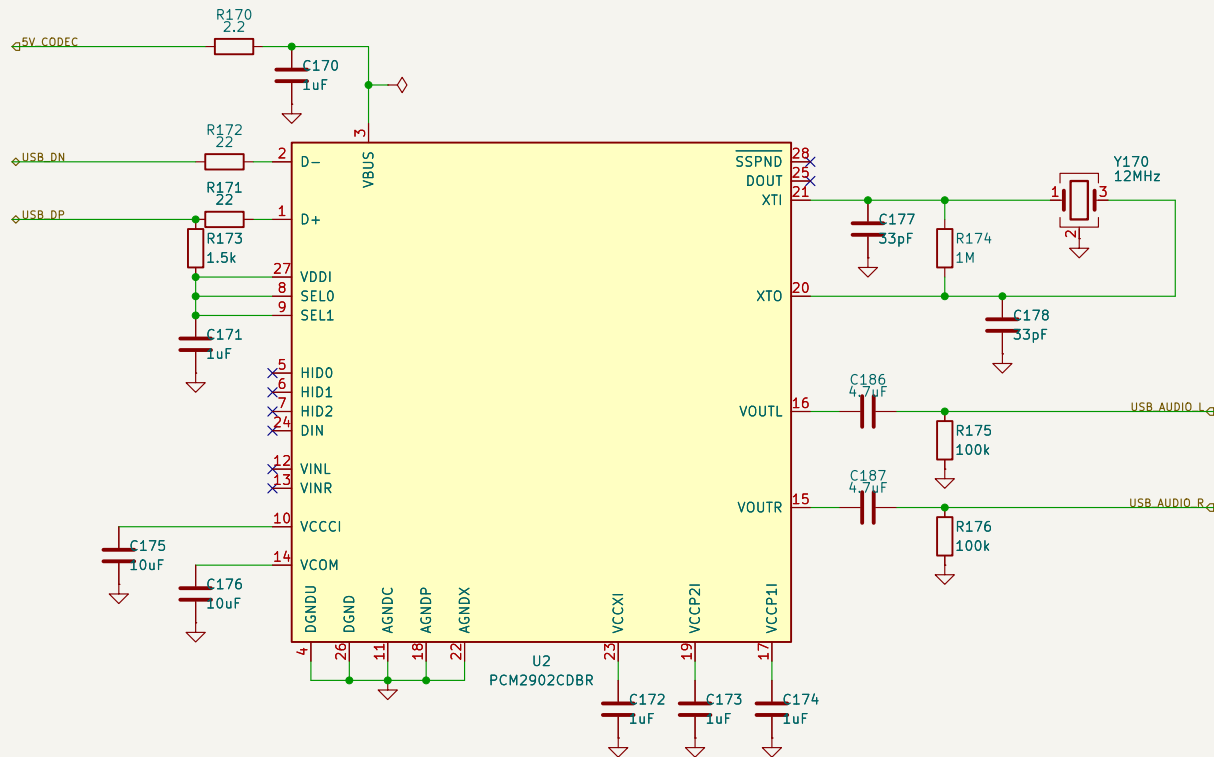
Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

Id: 7/12

Supply: 5V_CODEC (U23 load switch from 5V_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
 Codec is powered only when CODEC_PWR_EN (U23) and 5V_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC_PWR_EN on DCP/unknown sources.
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
 USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUDIO_L/R

Sheet: /USB_Audio/
 File: USB_Audio.kicad_sch

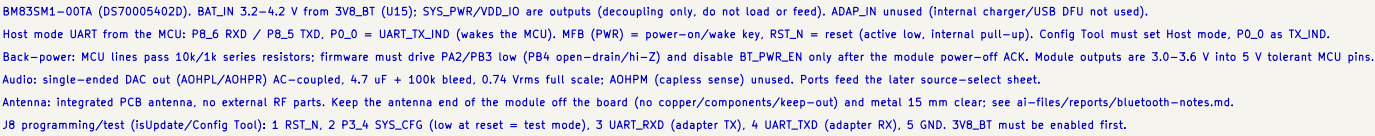
Title: USB audio

Size: A4
 KiCad E.D.A. 10.0.6-rc2

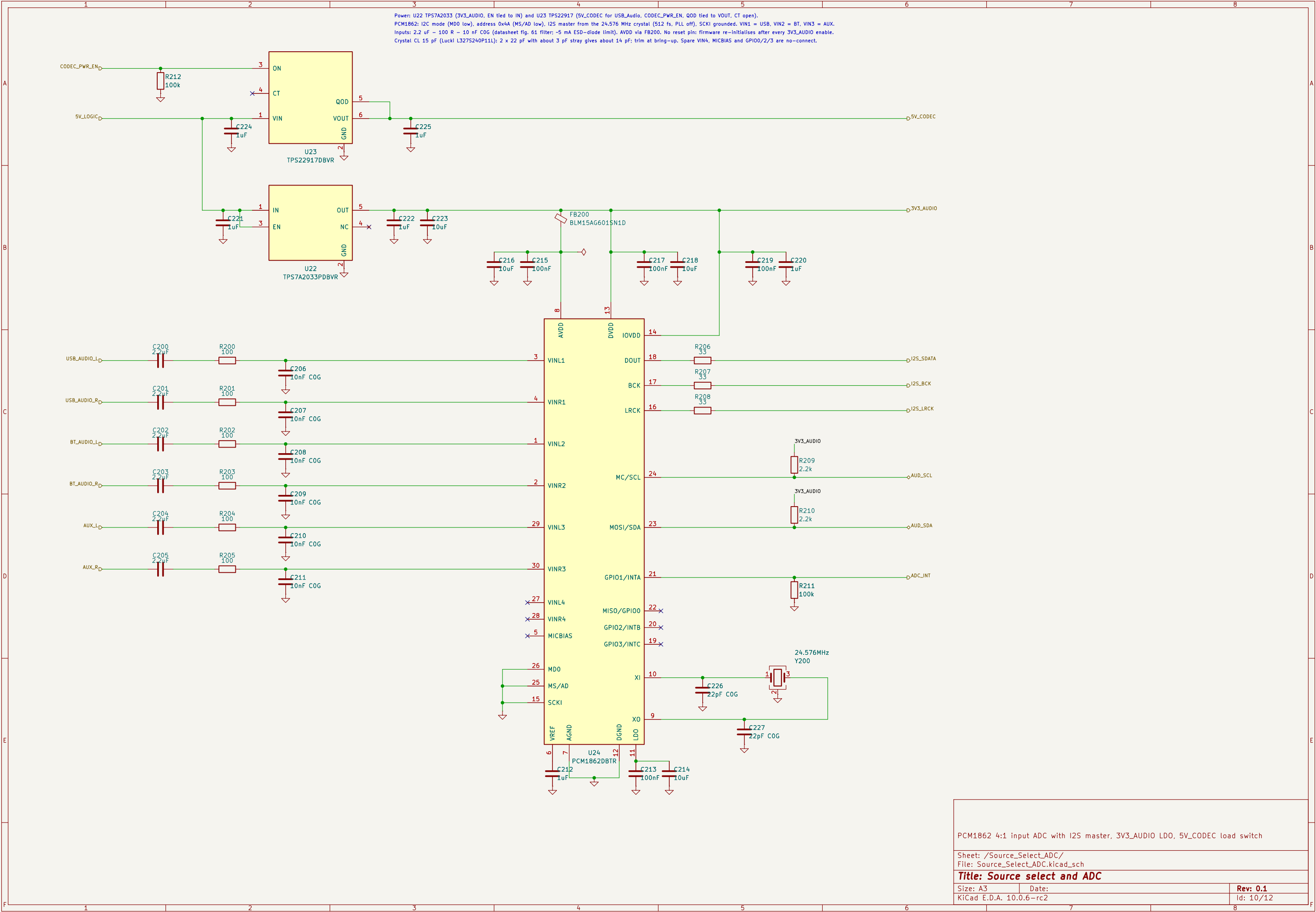
Date:

Rev: 0.1

Id: 8/12



Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/12

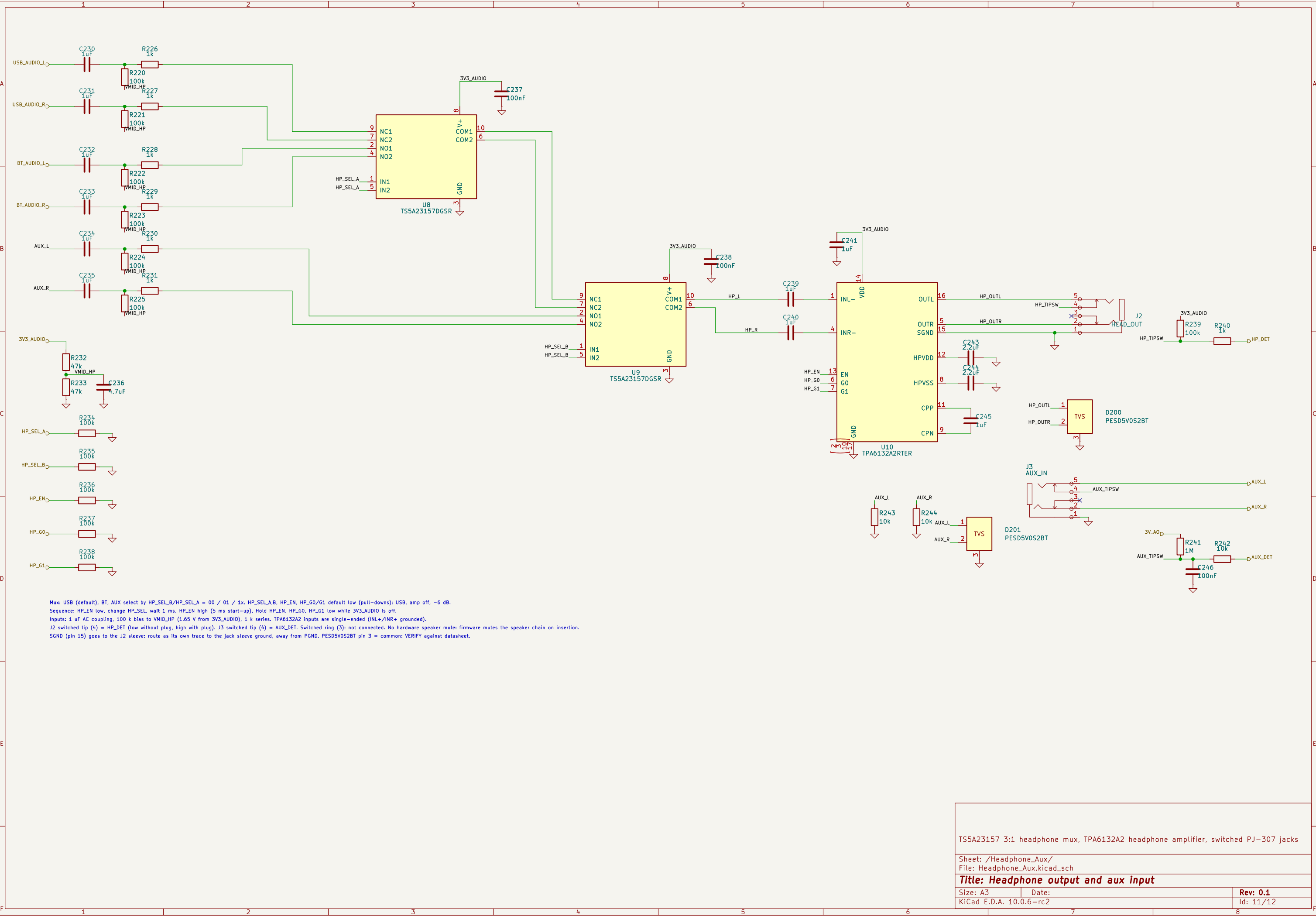


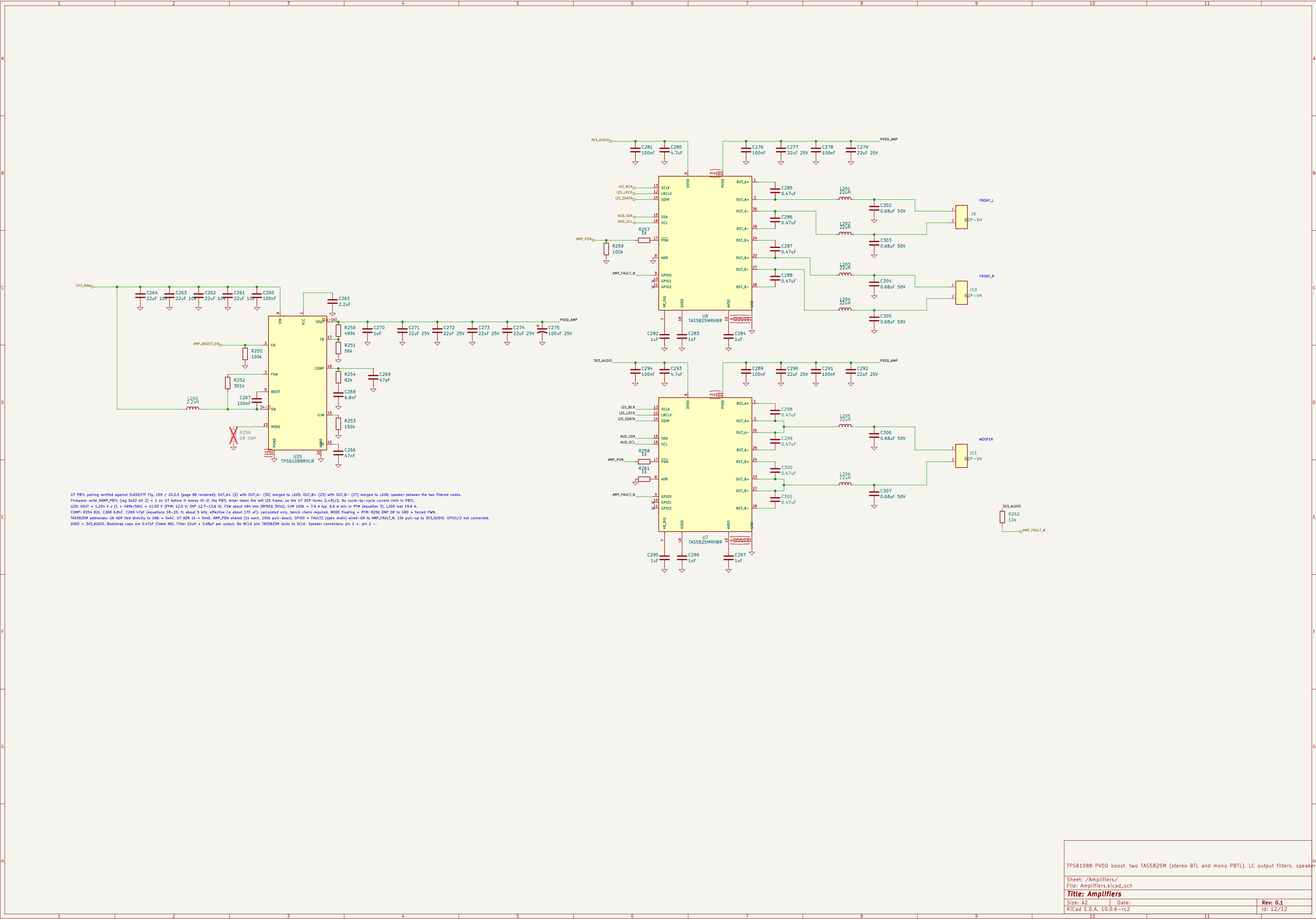
PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch

Sheet: /Source_Select_ADC/
File: Source_Select_ADC.kicad_sch

Title: Source select and ADC

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 10/12





U7 PBTl pairing verified against SLASEH7F Fig. 159 / 10.2.5 (page 86 rendered): OUT_A+ (2) with OUT_A- (30) merged to L205; OUT_B+ (23) with OUT_B- (27) merged to L206; speaker between the two filtered nodes.
Firmware: write DAMP_PBTl (reg 0x02 bit 2) = 1 on U7 before it leaves HI-Z; the PBTl mixer takes the left I2S frame, so the U7 DSP forms (L+R)/2. No cycle-by-cycle current limit in PBTl.
U25: VOUT = 1.204 V x (1 + 499k/56k) = 11.93 V (PFM: 12.0 V; OVP 12.7-13.6 V); FSW about 49k Hz (RFREQ 301k); ILIM 150k = 7.9 A typ, 6.6 A min in PFM (equation 3); L200 Isat 19.6 A.
COMP: R254 82k, C268 6.8nF, C269 47pF (equations 18-20, fc about 5 kHz; effective Co about 170 uF); calculated only; bench check required; MODE floating = PFM; R256 DNP OR to GND = forced PWM.
TAS5825M addresses: U6 ADR tied directly to GND = 0x4C; U7 ADR 1k = 0x4D; AMP_PDN shared (1k each, 150k pull-down); GPIO0 = FAULTZ (open drain) wired-OR to AMP_FAULT_N; 10k pull-up to 3V3_AUDIO; GPIO1/2 not connected.
DVDD = 3V3_AUDIO. Bootstrap caps are 0.47uF (Table 66). Filter 22uH + 0.68uF per output. No MCLK pin: TAS5825M locks to SCLK. Speaker connectors: pin 1 +, pin 2 -.